

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing **COURSE CODE:** CSA1506
and Big Data Analytics

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

Code of Conduct

I Ismail M (192524128)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ismail M

Assessment Tasks Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering.

Design a cloud architecture using scalability and caching techniques.

Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Ans:

1. Cloud Architecture Design (Scalability & Caching)

To deliver low-latency video streaming to millions of concurrent users, a multi-tier serverless and decoupled cloud architecture is required:

- **Content Delivery Network (CDN):** Deploy edge locations globally (e.g., AWS CloudFront / Cloudflare) to cache video segments (HLS/DASH) close to users, offloading 90%+ of traffic from the origin servers.
- **Storage Tier:** Store raw and transcoded video files in Object Storage (e.g., AWS S3), configured with appropriate lifecycle rules.
- **Compute Tier:** Use microservices hosted on containerized auto-scaling platforms (AWS ECS/EKS) or Serverless functions (AWS Lambda) for user authentication, recommendations, and stream initiation.
- **Caching Tier:** Use an In-Memory Datastore (e.g., Redis / Memcached) for fast session storage, user metadata, and video catalog caching.

2. Load Balancing & Auto-Scaling

- **Global Load Balancing:** Use DNS-based routing (e.g., Route 53) or Anycast IP to route user requests to the nearest healthy regional Application Load Balancer (ALB).
- **Application Load Balancers (ALB):** Distribute incoming traffic across microservice instances using round-robin or least-outstanding-requests algorithms.
- **Auto-Scaling Policies:**
 - **Predictive Scaling:** Scale up compute capacity automatically prior to scheduled events (e.g., prime-time shows or match streams).
 - **Dynamic Target Tracking:** Maintain CPU/Memory utilization around 60–70% or scale based on network throughput/request counts to dynamically absorb traffic spikes without buffering.

3. Cost Optimization Strategies

- **CDN Edge Caching:** Maximizing cache-hit ratios at the CDN level drastically reduces origin egress data transfer costs.
- **Tiered Storage Lifecycle:** Automatically move older, rarely watched videos from Standard S3 to S3 Infrequent Access (IA) or Glacier.
- **Spot & Reserved Instances:** Use Spot Instances for stateless video transcoding worker nodes and Reserved Instances/Savings Plans for predictable baseline workload services.

2. A fintech startup is deploying applications using containers across multiple cloud providers.

They face challenges in maintaining consistency and managing deployments.

Propose a solution using container orchestration and multi-cloud strategies.

Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Ans:

1. Proposed Solution

- **Container Orchestration:** Deploy a managed Kubernetes environment across cloud providers (e.g., AWS EKS, Google GKE) managed by a unified multi-cluster control plane (e.g., Red Hat OpenShift, Rancher, or Google Anthos).
- **Infrastructure as Code (IaC):** Use Terraform and Helm charts to provision declarative configurations identically across all cloud environments to ensure deployment consistency.

2. Service Discovery & Scaling Efficiency

- **Service Discovery:** Implement a Service Mesh (e.g., Istio / Linkerd) to manage inter-service communication, service registry, traffic routing, and mutual TLS (mTLS) security seamlessly across cloud borders.
- **Scaling:**
 - **Horizontal Pod Autoscaler (HPA):** Automatically adjusts container pod counts based on CPU, memory, or custom business metrics (e.g., transaction queue depth).
 - **Cluster Autoscaler:** Dynamically scales worker nodes up or down depending on unallocated pod resource demands.

3. Benefits and Risks of Multi-Cloud Model

Feature	Details
Benefits	High Availability & Redundancy: Prevents single-vendor lock-in and vendor outages. Compliance & Data Residency: Allows placing specific data in regions dictated by financial regulations.
Risks	Increased Complexity: Managing multiple cloud paradigms increases operational overhead. Data Transfer Costs: High inter-cloud egress fees when passing traffic between providers. Inconsistent Security Posture: Risk of misconfigurations across disparate provider IAM systems.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.

Provide steps to ensure business continuity during failures.

Ans:

1. Cloud-Native Backup, Replication, and Versioning

- **Versioning:** Enable Object Versioning on cloud storage buckets so accidentally deleted or overwritten objects can be restored to prior states instantly.
- **Automated Backups:** Implement automated snapshots for transactional databases (e.g., AWS RDS / Aurora) with cross-region automated backup copy tasks.
- **Cross-Region Replication (CRR):** Continuously replicate critical application data and database read-replicas asynchronously to a secondary geographical region.

2. Achieving RTO and RPO

- **Recovery Point Objective (RPO):** Set database write-ahead logging and transaction log backups every 5 minutes to limit potential data loss to under 5 minutes.
- **Recovery Time Objective (RTO):** Use an **Warm Standby** or **Pilot Light** disaster recovery model. Infrastructure in the secondary region is pre-configured via Infrastructure as Code (IaC), keeping restoration time under 15–30 minutes during a site outage.

3. Steps for Business Continuity During Failure

1. **Detection:** Automated monitoring (e.g., CloudWatch / Datadog) triggers an alarm upon detecting primary region failure.
 2. **Failover:** Route 53 DNS failover automatically shifts user traffic to the secondary region's endpoints.
 3. **Promotion:** Promote secondary read-replica databases to primary write mode.
 4. **Post-Recovery Synchronization:** Re-establish asynchronous data replication back to the primary region once restored.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements.

Design a hybrid architecture that ensures secure communication between environments.

Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Ans:

1. Hybrid Architecture & Secure Communication

- **Dedicated Connection:** Establish an encrypted private network link between the on-premises data center and the public cloud using AWS Direct Connect / Azure ExpressRoute combined with IPsec VPN for redundancy.
- **Data Segmentation:** Keep sensitive citizen data (PII) hosted on-premises or within a isolated Private Cloud, while non-sensitive applications leverage the public cloud.



2. Identity Management & Encryption

- **Identity Management:** Implement Federated Identity Management (FIM) using SAML 2.0 / OAuth2 with Active Directory / Azure AD to enforce Single Sign-On (SSO) and Role-Based Access Control (RBAC) across both environments.
- **Encryption at Rest:** Enforce AES-256 bit encryption across all databases and block storage using Hardware Security Modules (HSM) with customer-managed keys (KMS).
- **Encryption in Transit:** Mandatory TLS 1.3 protocol for all internal API communications and client-to-server traffic.

3. Key Challenges

- **Compliance & Auditability:** Ensuring continuous monitoring across hybrid boundaries to satisfy regulatory frameworks.
- **Network Latency:** Managing latency bottlenecks for applications requiring real-time synchronous database calls between cloud and on-premises environments.
- **Operations Overhead:** Maintaining dual monitoring tools, policies, and skill sets across legacy and cloud setups.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.

The system must continue operating even if one region goes down.

Propose a multi-region deployment strategy in the cloud.

Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Ans:

1. Multi-Region Deployment Strategy

- **Active-Active Setup:** Deploy identical stateless application environments in two or more geographically distinct cloud regions (e.g., us-east-1 and eu-central-1).
- **Traffic Routing:** Utilize Latency-Based or Geolocation-Based DNS routing to direct users to their nearest healthy active region.

2. Data Replication & Failover Mechanisms

- **Global Distributed Database:** Deploy multi-region active-active database clusters (e.g., Amazon DynamoDB Global Tables, Google Cloud Spanner, or CockroachDB) that synchronize writes across regions continuously.

- **Automated Failover:** Continuous health checks monitor the health of each region's load balancer. If an entire region degrades, DNS health checks mark it unhealthy and redirect 100% of incoming live traffic to the healthy standby regions instantly without human intervention.

3. Monitoring and Alerting Strategy

- **Synthetic Monitoring:** Run continuous automated browser scripts to test end-to-end user workflows (logins, payments) from global vantage points.
- **Centralized Telemetry:** Aggregate metrics, logs, and traces into a centralized platform (e.g., Prometheus/Grafana or Datadog).
- **Proactive Alerting:** Set threshold-based alerts (e.g., error rates > 1%, API response latency > 500ms) integrated with automated paging systems (PagerDuty) to notify on-call engineers prior to full-service degradation.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification

Clarity	15%	Clear and wellorganized response	Generally clear	Somewhat unclear	Poorly presented
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